

Waleed Khalid Alzamil

Google DeepMind Scholar | AI Researcher

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Summary

Incoming Google DeepMind Scholar at AIMS pursuing an MSc in AI for Science. Strong foundation in computer engineering, 3D geometric deep learning, and graph-based neural networks. Transitioning from applied ML engineering at ClearSky to academic research, with a focus on geometrical and topological deep learning. Aspiring to pursue a PhD in applied mathematics and AI to solve complex, planetary-scale structural problems.

Education

Master of Science (MSc) in AI for Science

African Institute for Mathematical Sciences (AIMS) 🔗

• **Funding & Honors:** Fully funded by the **Google DeepMind Scholarship**.

09/2026 – Present
Cape Town, South Africa

Bachelors of Engineering in Computer and System Engineering

Ain Shams University

• **Graduation Project :** AI-Automated Dental Crown Generation (Focus on 3D Point Clouds).

09/2020 – 07/2025
cairo, Egypt

Publications & Research Output

Peer-Reviewed Publication 🔗

"Between Text and Events: Process Context Graph (ProCG)" Accepted at ICPM 2026.

Mohammad Dallash, **Waleed Alzamil**, Kyrillos Eskander, Xixi Lu, Ahmed Elkholy

Research Collaborations & Acknowledgments 🔗

Noran Ayman M. Ismael, et al. "The Application of Artificial-based Models to Classify Oral Cavity Findings Based on Clinical Image Analysis". Advanced Dental Journal, Vol 7, Issue 4, 2025

Acknowledged for technical efforts in developing and refining the core AI classification model

Technical Reports & Preprints

- "AI-Automated Dental Crown Generation: 3D Teeth Segmentation and Generation". Graduation Project Technical Reports (Part 2), 2025. 🔗
- "AI-Automated Dental Crown Generation: 3D Teeth Segmentation and Generation". Graduation Project Technical Reports (Part 1), 2025. 🔗
- "Task Entities in Object-Centric Event Logs". ClearSky Internal Report, 2026. 🔗

Professional Experience

Machine Learning Engineer

Clear-sky.ai (Full time)

- Co-authored a paper on Process Context Graph (ProCG) accepted at ICPM 2026.
- Designed database schemas for scalable Object-Centric Process Mining (OCPM).
- Integrated with Outlook to trigger our system for all new emails in a shared mailbox using a webhook server protocol via Azure Graph API.
- Implemented MCP tools for OCPM.

06/2025 – 10/2026

AI Engineer

Tanweer (Part time)

- Leveraged ChatGPT to translate entire episodes, dramatically improving efficiency. This process reduced translation time **from 8 hours to just 30 seconds** and eliminated traditional translation costs. This System has been deployed on GCP for the company.
- Notice some improvement in fine-tuned LLMs such as RWKV-4-World LLM for English-to-Arabic and Arabic-to-English translation.

07/2023 – 06/2024
cairo, Egypt

NLP Engineer

Siemens (Internship)

- Code Coverage Enhancement: Explored the enhancement of code coverage by integrating Large Language Models (LLMs) into test case generation.
- Coverage Metrics: Utilized GCOV to compute code coverage, creating a curated database of codes, test cases, and coverage metrics.

09/2023 – 01/2024
cairo, Egypt

Projects

AI-Automated Dental Crown Generation

Graduation Project

- Implemented and experimented with various architectures, including FoldingNet, DGCNN, PCT and Transformer-based models.
- Successfully trained a segmentation model and achieved **95% Accuracy**.
- Successfully trained a model with **contrastive learning** to preserve any upper or lower arch in a fixed canonical space, more details in Crown-Generation part 02.
- Segmentation Report 🔗

09/2024 – 06/2025

- Docker-hub segmentation model [↗](#) Deployment Repo [↗](#) Development Repo [↗](#)

3D Interactive Virtual Estate (3DIVE) [↗](#)

11/2023 – 06/2024

Aims to enhance user engagement through immersive property exploration

- Created a full AI pipeline for extracting **frames from videos**, estimating **camera poses with COLMAP**, and reconstructing **3D models** using advanced algorithms (**Gaussian Splatting** and **SuGaR**).

Othello Game [↗](#)

04/2024 – 06/2024

- Implemented the rules and heuristics of the Othello game and developed a user-friendly GUI.
- Implemented Minimax and Alpha-Beta Minimax algorithms.
- Developed **Actor-Critic RL algorithms** to compete against the Minimax algorithms.
- Enabled computer vs. computer gameplay (RL vs. Minimax).

Synergy EMGs Proportional Control [↗](#)

07/2023 – 08/2023

Using Electromyography data for non-invasive naturally controlled robotic hand prostheses

- Collected data using EMG sensors and Qualisys software.
- Configured cameras and sensors, and prepared the environment for data collection, including calibration.
- Trained different architectures under different assumptions using the **Ninapro dataset** for proportional control to predict the angles of defined joints.

Volunteering

Ain Shams University Racing Team (ASURT)

05/2023 – 05/2026

Deep Learning Team Lead

cairo, Egypt

- **Collaborative Research Project** [↗](#) : Partnered with a dental PhD student to develop deep-learning models for classifying variational dental cases, resulting in a publication in the Advanced Dental Journal.
- Mentored team members on advanced deep learning topics and led the perception module development for Formula Student.
- Designed and implemented a RAG test case generator to enhance the AI model.

ASURT with collaboration STP

11/2022 – 02/2023

Technical Team Member Macathon 4.0

cairo, Egypt

Developed robotics algorithms in Python and C++ for tasks including perception, decision-making, and control of autonomous cars.

Ain Shams University Racing Team (ASURT)

08/2022 – 05/2023

Formula 10th Team Member

cairo, Egypt

Implemented global and local planning algorithms for an autonomous racing vehicle. Completed a 2-month intensive workshop covering ROS, LIDAR perception, localization, path planning, and navigation control.

Skills

Research Interests — Geometrical Deep Learning, Topological Deep Learning, Graph Neural Networks, Applied Mathematics

Machine Learning — PyTorch, TensorFlow, Scikit-learn, Huggingface, DSPy, AutoGen | **3D & Geometric Vision** — Point Cloud Processing (DGCNN, PCT), Gaussian Splatting, Open-CV | **Engineering** — Python, C++, Linux, Docker, PostgreSQL, FastAPI, ROS, Git